Project title: 48-Hour Day-Ahead Electricity Price Forecasting in ERCOT

Group:
Jordan Hernandez-Mares
Uniqname: jordh
UMID: 58132290

Kevin Farokhrouz
Uniqname: krouz
UMID: 67469130

Shreya Nadiger
Uniqname: snadiger
UMID: 66211386

Problem description

Motivation:

Wholesale electricity prices in Texas's ERCOT market are among the most volatile in the United States. Hourly prices are usually between $20 and $60 per MWh but can rise to thousands of dollars during heat waves, cold snaps, or periods of low wind and solar output. ERCOT publishes forecasts of demand, wind, and solar generation, but it does not publish price forecasts. Many market participants need to know which hours will be expensive. Retail electricity providers buy power on the wholesale market and sell it to customers at fixed rates, so unexpected price spikes cause direct losses. Battery operators earn money by charging when prices are low and discharging when they are high. Large flexible users such as data centers and industrial plants can shift or reduce their usage during expensive hours. Better price forecasts also help batteries and flexible users move demand toward hours when cheap solar and wind power are available.

Upon evaluating results, we look to answer the following questions:

- Does our model achieve lower forecast error than the naive and LEAR baselines?
- How much does accuracy drop between forecasts for tomorrow and the day after tomorrow?
- How does accuracy during price spikes, defined as hours above the 95th percentile of prices, compare with accuracy during normal hours?
- Do our prediction intervals contain the actual price about 80% of the time?

Concrete Problem Statement:

Input: Historical hourly electricity prices from one ERCOT trading hub (Historical DAM Load Zone and Hub Prices), forecasts of electricity demand and wind and solar generation, net load (demand minus wind and solar generation), recent natural gas prices, and calendar features such as hour, weekday, and holidays. All inputs will be restricted to information available before the daily 9 AM forecast time to prevent future information from entering the model.

Output: A forecast of 48 hourly day-ahead electricity prices for the following two days, with a 10th–90th percentile prediction interval for each hour.

For example, a forecast made at 9 AM on a Monday in August would produce 48 hourly prices for Tuesday and Wednesday. The model might predict prices near $25/MWh around noon on Tuesday, when solar output is high, and rising to several hundred dollars around 8 PM, when solar output falls while air conditioning demand stays high. Once both days have passed, we compare each predicted hour against the actual day-ahead price.

Method: Our starting model will be a tree model (like LightGBM) trained on the features above, using quantile regression to produce the 10th, 50th, and 90th percentile forecasts for each hour. If time allows, we will compare it with a small neural network trained on the same features.

Related work and our contribution:

Lago et al. (2021) introduced an open benchmark for day-ahead electricity price forecasting covering several European markets and PJM in the United States, but not ERCOT. Other research, such as Marcjasz et al. (2023), examines uncertainty in electricity price forecasts.

Our project will focus on two aspects. First, we will evaluate electricity price forecasting in ERCOT, extending the evaluation beyond the markets included in the Lago et al. benchmark. Second, we will forecast prices for the following two days and report results separately for each day. This will allow us to compare how accurately we predict prices for tomorrow and the day after tomorrow, and understand how forecasting further ahead affects accuracy.

Evaluation plan:

Dataset:

We will evaluate hourly electricity price forecasts for the following two days at one ERCOT trading hub. We will build a dataset from public ERCOT data covering 2019 through 2025, including historical day-ahead prices, ERCOT's archived demand and renewable generation forecasts, natural gas prices, and calendar information. We will train on 2019 through 2023, validate on 2024, and test on 2025. Inputs will include only information available at the time each forecast would have been made. All metrics will be reported separately for tomorrow and the day after tomorrow, and separately for spike hours and normal hours.

Primary success metrics:

1. Mean Absolute Error (MAE): The average absolute difference between predicted and actual electricity prices, measured in dollars per MWh. This will be our main measure of forecasting accuracy.
2. Root Mean Squared Error (RMSE): A measure that penalizes large prediction errors more heavily, helping assess performance during price spikes.
3. Prediction interval coverage: The percentage of actual prices that fall inside our 10th to 90th percentile range. A well-calibrated model should capture about 80% of actual prices. We will also report the average width of the interval, since a very wide interval can reach 80% coverage without being useful.

Baselines:

We will compare our forecasts against simple methods that predict each hour's price using either the same hour on the previous day or the same hour one week earlier. We will also compare against LEAR, an established electricity price forecasting approach from Lago et al. (2021).